

Exercise 2

A neutrino produced by a left-chiral charged-current interaction is not a pure helicity eigenstate if $m_\nu \neq 0$.

- ★ Show that for $E_\nu \gg m_\nu$, the probability of the opposite-helicity component is approximately

$$P_{\text{opp}} \simeq \left(\frac{m_\nu}{2E_\nu} \right)^2.$$

- ★ Using $E_\nu = 1 \text{ MeV}$ and $m_\nu < 0.45 \text{ eV}$, estimate P_{opp}
- ★ Explain why this makes helicity and chirality effectively equivalent for relativistic neutrinos.

Exercise 3

- ★ Consider electron capture

$$A(J = 0) + e^- \rightarrow B^*(J = 1) + \nu_e,$$

where the captured electron is in an s -state. Choose the z -axis along the neutrino momentum.

- List the allowed spin projections of the neutrino and of B^* .
- Show that the spin projection $S_z(B^*)$ tags the neutrino helicity.
- Explain why this statement becomes ambiguous if the quantization axis is chosen independently of the neutrino momentum.